When is an engineering check sufficient?

Claim-dependent observability, numerical qualification and retained counterexamples

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Engineering research report. Source snapshot: `da661f54fcd7`; retained portfolio pin: `0381c1d2bfe6`. Synthetic development study and preserved FEA; no physical validation or completed reviewer-policy benchmark.

Abstract

An engineering calculation can finish successfully and match a familiar stress result while still supplying insufficient evidence for another claim. This project makes that distinction explicit using finite sets of implementation explanations, exact reference problems and separately assembled numerical checks. In a 100 mm axial bar under prescribed force, halving modulus leaves stress at 50 MPa but doubles extension from 0.025 to 0.050 mm; stress agreement therefore cannot resolve the extension claim. Under prescribed displacement, the same fault changes stress and reaction, illustrating that check value depends on boundary conditions. A retained development sweep covers 878 parameter variants across bending, pressure-only cylinder and layered-wall families. Ninety-six of 320 cylinder variants fail inherited numerical gates; a subsequent registered refinement investigation qualifies 99 inputs at unchanged budgets while retaining the original failures. Historical mounting-plate FEA completes and agrees with its chosen reference but does not establish mesh convergence. Current review records admit five scoped development base problems, including one firm published-reference problem, through non-blind AI technical review. No cross-family frozen labels or scored reviewer-selection benchmark exists. The report answers a bounded question: evidence is sufficient only relative to a named claim, allowed explanations, observation model and artifact class; neither solver completion nor one passing observable is a general certificate.

Completed scope. This report closes the retained analytical and numerical development study at the source revisions named below. Its scope is five admitted reference problems, the seeded observability examples, the 878-variant sweep, the registered cylinder-refinement investigation and the preserved mounting-plate FEA. The planned cross-family reviewer-policy benchmark was not executed. Closing this development scope neither completes that benchmark nor changes the retained fail/unknown labels. No physical artifact was validated.

1 Question, hypothesis and contribution

Central question. Can a small set of engineering checks establish that a computed result is trustworthy, and when should a reviewer abstain? The working hypothesis is that sufficiency is claim-dependent: a check is useful only when it separates allowed explanations that disagree about the correctness of the claim. It need not identify the exact implementation fault, but it must resolve the decision that the evidence is supposed to support.

Answer supported by this study. A fixed stress check is insufficient for the force-controlled bar's deflection claim even though stress itself is correct. Boundary conditions change which outputs expose a material or geometry error. Numerical reference agreement also fails to establish mesh stability. These examples establish concrete counterexamples to stronger assurance shortcuts; they do not establish that a learned reviewer or any proposed policy is generally better.

The portfolio records the owner's work as formulating the claim/evidence contract, assembling exact references and independent computational routes, implementing seeded artifacts and check policies, preserving FEA records, and designing the development sweeps. Development and documentation were

AI-assisted. The source separately identifies Claude AI technical reviews, their exposed inputs, scripts and reconciliation. These were non-blind and were adopted for scoped development admissions; they are not external peer review, independent human review, physical validation or a scored LLM-reviewer experiment. This LaTeX report is a Codex-assisted synthesis of retained artifacts, with no new numerical campaign.

1.1 Relation to established work

Equivalence-class determination already formalizes selecting tests to distinguish decision classes rather than individual hypotheses [2]. Code verification also has established independent-reference and manufactured-solution methods [3]. This project does not claim to invent either idea. Its proposed contribution is to operationalize the distinction for engineering evidence: preserve claims, explanations, observation tolerances, check costs and provenance, so that apparently successful checks can be challenged by explicit witnesses. The repository is still building the reference and labeling foundation needed to evaluate that proposal.

2 Claim correctness versus evidence sufficiency

Let $\mathcal{H}$ be a declared finite set of implementation explanations. A claim c has a reference q_c^* and a numerical acceptance tolerance t_c; its correctness label for explanation h is

$$L_c(h) = \mathbf{1}\{|q_c(h) - q_c^*| \le t_c\}. \tag{1}$$

Given observed check outputs $D = \{(j, y_j, b_j)\}$, retain explanations compatible with the observation tolerances:

$$\mathcal{H}_D = \{h \in \mathcal{H} : |q_j(h) - y_j| \le b_j \text{ for each observed } j\}. \tag{2}$$

Evidence resolves claim c if $\mathcal{H}_D$ is nonempty and L_c is constant over it. All labels true resolves acceptance; all false resolves rejection. An empty compatible set means inconsistent evidence, never automatic sufficiency. Mixed labels give an explicit opposite-decision witness and require abstention or another check.

This definition is conditional on the declared explanation set. An unlisted unit error, boundary-condition mistake or nonlinear mechanism can invalidate an otherwise sufficient finite-set decision. The claim tolerance in Eq. (1) is a numerical comparison rule, not an allowable stress or a physical design margin. Likewise, observation compatibility and claim correctness are separate: fitting a check output need not imply that the named claim is right.

3 Axial-bar experiment: a transparent observability test

3.1 Model, protocol and controls

The model is a uniform, linearly elastic axial bar, fixed at one end, with length $L = 100$ mm, area $A = 20$ mm^2, and nominal modulus $E = 200000$ MPa. The reference tuple is stress, end displacement and positive tensile reaction magnitude. A closed-form route is cross-checked with a separately assembled stiffness model. The two loading conventions are configurations of one family, not independent samples.

For prescribed tensile force F,

$$\sigma = F/A, \qquad \delta = FL/(EA), \qquad R = F. \tag{3}$$

For prescribed end displacement δ_0,

$$\sigma = E\delta_0/L, \qquad R = EA\delta_0/L, \qquad \delta = \delta_0. \tag{4}$$

The nominal settings $F = 1000$ N and $\delta_0 = 0.025$ mm give the same tuple: 50 MPa, 0.025 mm and 1000 N. Artifacts include a correct control, benign mesh partition, nuisance density change, half modulus, half area, and changed loading. Completion checks only that outputs are finite. Numerical comparisons use stored unrounded values.

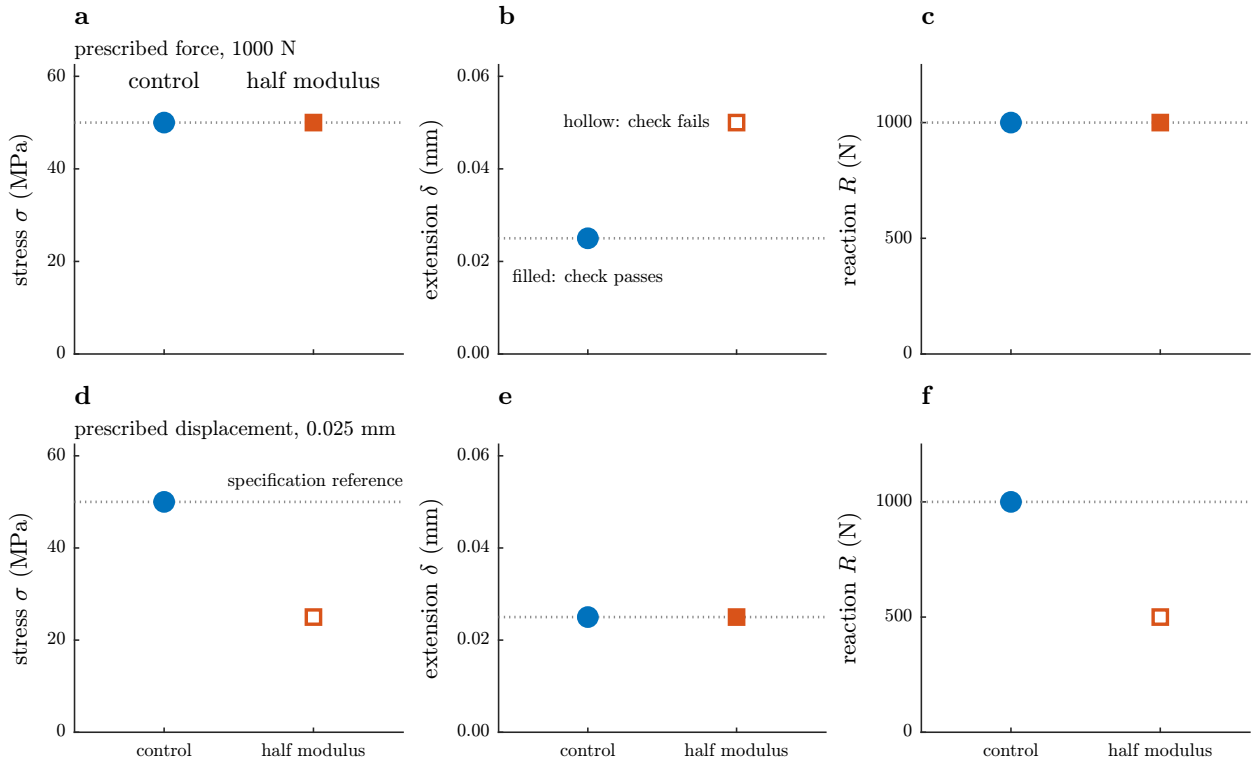


Figure 1: Project-created six-panel result figure from retained axial-bar data. Blue is the nominal control; orange is half modulus. Filled markers pass their observable check; hollow markers fail. Modulus is invisible to force-controlled stress and reaction, but visible to prescribed-displacement stress and reaction.

Table 1: All seeded axial artifact outputs. C/B/N denotes correct, benign-partition and nuisance-density controls, each with identical values. Full tuple correctness is separate from any individual passing output. Source: evidence-sufficiency matrix and JSON.

Loading	Artifact	σ (MPa)	δ (mm)	R (N)	Tuple
Force	C/B/N (3 artifacts)	50	0.0250	1000	Correct
Force	Half modulus	50	0.0500	1000	Wrong
Force	Half area	100	0.0500	1000	Wrong
Force	Loading fault	25	0.0125	500	Wrong
Displacement	C/B/N (3 artifacts)	50	0.0250	1000	Correct
Displacement	Half modulus	25	0.0250	500	Wrong
Displacement	Half area	50	0.0250	500	Wrong
Displacement	Loading fault	25	0.0125	500	Wrong

3.2 Finite witness and causal interpretation

For the claim-specific witness, the explanation set contains only nominal modulus and half modulus, with all stated geometry, loads and assumptions fixed. Stress of 50 MPa observed within 10^{-8} MPa is compatible with both explanations under force control. Both explanations have correct stress, so stress is resolved as acceptable. One predicts correct deflection and one does not, so deflection is unresolved. With

prescribed displacement, only nominal modulus is compatible with observed 50 MPa stress, resolving both claims within this finite set. The deflection correctness tolerance is 10^{-8} mm.

This is not merely a small signal lost in noise. Under force control, $\partial\sigma/\partial E = 0$ exactly in this model. No more precise stress instrument would reveal the modulus fault. Under displacement control, stress depends directly on E while displacement is imposed. A check-selection rule must therefore know the boundary condition and the claim, not just the output name or mutation identifier.

Table 2: Comparator smoke test on the same twelve seeded development artifacts. Random results are exact expectations over equal-cost choices, not sampled trials. Costs are abstract acquisition tokens, not measured engineering labor.

Comparator	Wrong accepts	Abstain	Coverage (%)	Cost/artifact
Fixed stress	2	0	100	1
Random (exact expectation)	3	0	100	1
Modulus sensitivity	1	0	100	1
All checks	0	0	100	4
Cautious stress	0	8	33.3	1
Always abstain	0	12	0	0

Wrong rejects are zero in this selected development set. The table demonstrates error/coverage/cost tradeoffs, but cannot rank general policies: the artifacts were seeded, the family is public, costs are abstract, and no LLM reviewer was run. A policy can avoid every false acceptance simply by abstaining always, making coverage indispensable to interpretation.

4 Reference families and independent computational routes

The expanded reference suite introduces distinct mechanisms rather than calling every parameter setting a new problem. Current admissions include five base problems at explicitly limited scopes. Only the pressure-only VM25 cylinder counts as a firm published-reference problem. Coverage remains 5/10 against a provisional target and 1/2 on the published-reference target.

4.1 Distributed-load cantilever

The bending model is Euler–Bernoulli with a uniform dead load, ideal clamp and nominal section stress. For load per length w, length L, second moment I, section extreme-fiber distance c and modulus E,

$$\delta_{tip} = \frac{wL^4}{8EI}, \quad |\sigma_{root}| = \frac{|w|L^2 c}{2I}, \quad R = -wL, \quad M = -wL^2/2. \tag{5}$$

Exact rational references are checked against consistent-load stiffness solutions. The nominal retained answers are 0.000486 m, 5.4 MPa, −6 N and −1.8 N m. Zero/reversed load and double modulus controls check sign, limiting behavior and observability. Root nominal stress is independent of modulus, but tip displacement is not. Shear deformation, local clamp stress, self-weight, geometric nonlinearity and material nonlinearity are excluded.

4.2 Pressure-only long cylinder

For inner radius a, outer radius b, internal/external pressure p_i, p_o,

$$A_c = \frac{p_i a^2 - p_o b^2}{b^2 - a^2}, \qquad B_c = \frac{a^2 b^2 (p_i - p_o)}{b^2 - a^2}, \tag{6}$$

$$\sigma_r(r) = A_c - B_c/r^2, \qquad \sigma_\theta(r) = A_c + B_c/r^2, \tag{7}$$

$$u_r(r) = \frac{(1-\nu)A_c r + (1+\nu)B_c/r}{E}. \tag{8}$$

The declared convention has zero axial end resultant and free uniform axial strain; plane strain or closed-end pressure would give a different displacement answer. Rotation in the full vendor VM25 example is excluded. Exact formulas are compared with a radial boundary-value solution integrated by RK4, plus separate traction, equilibrium and energy/work checks. The official verification case supplies published target context [4]; its rounded display is not the numerical solver budget.

Table 3: Nominal pressure-only VM25-derived reference, $a = 4$ in, $b = 8$ in, $p_i = 30000$ psi, $p_o = 0$, $E = 30 \times 10^6$ psi, $\nu = 0.3$. Source: qualification matrix.

Observable	Exact-derived value	Claim budget
Inner radial displacement (in)	0.00786666667	4×10^{-9}
Inner radial stress (psi)	−30000	0.03
Inner hoop stress (psi)	50000	0.03
Mid-wall radial stress (psi)	−7777.77778	0.03
Mid-wall hoop stress (psi)	27777.7778	0.03
Outer radial stress (psi)	0	0.03
Outer hoop stress (psi)	20000	0.03

4.3 Layered steady conduction

The wall has perfect layer contact, constant properties, no heat generation and film convection at both sides. In consistent units, with area A_w,

$$R_{th} = \frac{1}{h_i A_w} + \sum_j \frac{L_j}{k_j A_w} + \frac{1}{h_o A_w}, \qquad Q = \frac{T_f - T_a}{R_{th}}, \quad T_o = T_a + \frac{Q}{h_o A_w}. \tag{9}$$

A separately assembled heat-balance route checks exact-input references. A Dirichlet slab control gives midpoint 350 K and flux 100 W/m^2; doubling conductivity changes flux to 200 W/m^2 without changing midpoint temperature. The VM92-parameter wall uses vendor geometry/properties [5], but the printed outer-face target is disputed and excluded from published corroboration. The exact equations remain unchanged. Heat rate and inner-face displays supply only coarse support; the derived problem is not counted as a second firm published reference.

4.4 Kirsch exact restriction

The fifth admitted problem is the exact restriction of an infinite-plane, remote-y Kirsch field to a quarter-square domain outside a circular hole. It is not a uniformly loaded finite plate. Position-dependent full traction, including shear, is applied on both artificial outer boundaries. Hole traction is zero; normal symmetry displacements remove rigid modes. The OpenGeoSys benchmark supplies the equation/loading convention [6].

An independent code route takes Cartesian second derivatives of the Airy potential

$$\Phi = \frac{S}{4}\left[r^2 - 2a^2 \ln(r/a) + (r^2 - 2a^2 + a^4/r^2)\cos 2\theta\right], \tag{10}$$

using $\sigma_{xx} = \Phi_{yy}$, $\sigma_{yy} = \Phi_{xx}$ and $\sigma_{xy} = -\Phi_{xy}$. Agreement with a polar-stress route, traction, constitutive compatibility and boundary resultants are discriminating implementation checks. Smooth-potential equilibrium alone cannot establish correctness. The hole's $+x$ hoop claims are 30 MPa for nominal and scaled geometry, -21 MPa for reversed load, and zero for zero load. The original all-level boundary-quadrature hypothesis failed. A registered postpilot amendment retained those failures, added finer levels, and changed the rule to finest-level closure at unchanged budgets. Its later pass is not confirmation of the original all-level hypothesis. No FE solve or mesh-convergence claim is attached to this reference.

5 Expanded development experiment

5.1 Protocol and immutable failures

The retained deterministic sweep varies parameters within registered linear model families and runs separate reference/numerical routes under inherited tolerances. It uses one worker and one numerical thread; random sampling and a Monte Carlo seed do not apply. Bending and wall checks require exact rational agreement on inherited partitions. Cylinder claim tolerances are numerical budgets, separate from quadrature balance residuals. The run records source hashes, execution metadata, checkpoint and raw-artifact hashes; total runtime is 2.737 s with recorded peak RSS 42.73 MiB.

Table 4: Full retained reference sweep, not independent physical specimens. Source: development-compute matrix/JSON.

Family	Variants	Qualified	Failed gate	Comparisons	Failed comparisons
Cantilever	72	72	0	864	0
Pressure-only cylinder	320	224	96	6720	372
Layered wall	486	486	0	5832	0
Total	878	782	96	13416	372

The first failed cylinder has $a = 1$ in, $b = 4$ in, $E = 30 \times 10^6$ psi, $\nu = -0.5$, $p_i = 300$ psi, $p_o = 0$. Its coarse result was not relabeled after adding diagnostic finer steps. Negative Poisson ratio is a declared synthetic parameter, not an assertion that a particular physical specimen has that property. At the worst retained input, error/budget decreases from 122.784 at 32 steps to 7.76653, 0.483114, 0.0300368 and 0.00187103 at 64, 128, 256 and 512 steps. High-precision arithmetic retains the coarse error, supporting under-resolution rather than roundoff as the mechanism.

5.2 Focused cylinder refinement

A subsequent prospectively specified investigation covers the 96 earlier failures and three controls. Resolution is selected using successive numerical outputs and independent balances before the exact reference is consulted for final comparison. All 99 inputs qualify at unchanged budgets: 12 failures and three controls at 128 steps, 36 failures at 256, and 48 at 512. Observed non-roundoff-limited order spans 3.998735–4.010016. The maximum selected claim error is 0.00669941 times its original budget.

This qualifies numerical resolution only for the declared equations/inputs. The safety-factored step-difference rule is conditional on observed asymptotic behavior; it is not a certified universal error bound. The fourth-order behavior of this one-dimensional integrator cannot establish convergence of the historical unstructured FE model.

5.3 Detectability under bounded observations

The fault experiment applies a 1% agreement tolerance on a declared output scale and bounded observation error of 0.2%. Cylinder observations also include their verified numerical budget. Wall temperature

Table 5: Worst selected independent cylinder residuals or quadrature estimates, normalized by their declared characteristic scales. Every allocation is 2.5×10^{-7}. Source: cylinder-convergence matrix.

Check	Largest selected normalized value
Equilibrium residual	4.83945×10^{-8}
Energy/work residual	1.27186×10^{-7}
Equilibrium quadrature estimate	8.50487×10^{-8}
Energy quadrature estimate	2.26229×10^{-7}
Inner traction	0
Outer traction	1.21266×10^{-16}

is the outer-surface rise above ambient, avoiding the meaningless percentage change of an affine absolute temperature. All nominal/changed specimens used for this test passed their own numerical gates; earlier failed sweep inputs are excluded.

Table 6: Conditional detectability boundaries, rounded from retained bisection brackets. Reduction refers to modulus or layer-2 conductivity. The intermediate region depends on bounded observation error, not a probability.

Observable	Exit guaranteed pass (%)	Enter guaranteed detection (%)
Beam root stress	None in domain	None in domain
Beam tip displacement	0.793651–0.793654	1.185769–1.185772
Cylinder inner hoop stress	None in domain	None in domain
Cylinder inner displacement	0.793600–0.793603	1.185820–1.185822
Wall heat rate	1.088819–1.088822	1.630855–1.630858
Wall outer rise	1.088819–1.088822	1.630855–1.630858

The null spaces and below-tolerance signals are distinct outcomes. Root stress and cylinder hoop stress do not depend on modulus under the specified traction loads. Their absence of a crossing is structural. Deflection changes continuously with modulus, yet small changes remain indistinguishable at the chosen observation allowance. Neither phenomenon should be reported as a numerical solver failure.

6 Historical FEA: completion is not convergence

The retained mounting-plate records are historical numerical solves, not a new host run. Four global meshes complete and agree with their chosen reference, yet the last peak-stress change is 7.33% against a retrospective 1% stability limit. Two local hole-refinement records contain one mesh each, so their stability is unknown. Unknown is not converged.

Richardson extrapolation and a grid-convergence index are unavailable here. Successive peak-stress changes reverse sign, failing a monotone leading-error interpretation. For displacement, the finest-three change ratio is 0.259934; with the recorded target sizes a positive leading-error order would require more than 1.40942. The local records also change hole divisions from 24 to 48 while the global size coarsens from 3 to 5 mm. They are not one systematic global-refinement sequence. Deriving an effective mesh spacing solely from element counts or fitting an oscillating peak would conceal those differences.

This is a second independent answer to the central question: reference agreement and solver completion do not supply the missing discretization evidence. The reference itself may be appropriate for one nominal observable while the local peak remains mesh-sensitive. The retained fail/unknown labels are therefore evidence, rather than inconvenient bookkeeping to remove.

7 Review, display precision and threats to validity

The current case-admission ledger supersedes older zero-admission status wording while preserving its numeric sources. Five base problems are admitted at their scoped development claims, not as physical models or frozen policy-test labels. The reviewer had seen specifications, code and published targets; the separate script imported no repository reference routine but was written by an exposed AI reviewer. A later Kirsch review has its own pinned provenance. These records are technical cross-checks, not independent authorship or a blinded performance study.

Published displays are a further evidence class. The official VM25 displacement target 0.0078666 in differs from the exact-derived 0.007866666667 by 6.67×10^{-8} in: outside half a final displayed unit, inside one unit. The published rounding convention cannot be guessed into a tighter acceptance band. VM92 explicitly labels rounded-off values, yet the printed outer-face target remains excluded from corroboration under the project record. A coarse printed number, exact calculation and a numerically bounded solver result require different compatibility rules; none can silently borrow another's precision.

Internal validity is limited by shared problem assumptions, exposure during review and finite explanation sets. External validity is limited by linear toy families, selected faults and absent physical measurements. Comparison counts are not independent evidence samples. Abstract token costs omit setup, engineering effort and context acquisition. No cross-family truth-label release, applicable-check/cost matrix or scored reviewer-policy campaign has been completed. Consequently the study supports methodology and specific falsifying examples, rather than a finished benchmark ranking humans, LLMs or algorithms.

8 Conclusions

1. A passing force-controlled stress check does not establish the same bar's deflection claim: halving modulus preserves 50 MPa stress while doubling extension from 0.025 mm to 0.050 mm (Eqs. 3–4, Fig. 1, Table 1).

2. Five base problems are admitted at scoped development claims; only the pressure-only VM25 cylinder supplies a firm published-reference problem (Section 4). These counts replace the older four-problem summary without altering the earlier result files.

3. The 878-variant development sweep retained 96 failed cylinder gates. The subsequent registered investigation qualified 99 failure/control inputs at unchanged budgets; it did not retroactively turn the original coarse results into passes (Sections 5 and 5.2).

4. Historical mounting-plate solver completion and reference agreement did not establish mesh stability: the final peak-stress change was 7.33 %, and the local records had only one mesh each (Section 6).

5. The work supplies explicit observability counterexamples and qualified development references. It does not supply frozen cross-family policy-test labels or a completed reviewer-selection benchmark (Section 7). The design recommendation is to select checks against the named claim and unresolved alternative explanations, rather than treating a familiar output as a general certificate.

9 Future work

1. Run the lowest-cost discriminating experiment first: a small blinded accept/reject/abstain pilot using frozen retained artifacts, declared truth labels and equal-cost evidence choices. Report coverage, false acceptance, false rejection and acquisition cost together.

2. Extend the reference suite only after declaring the new artifact class, admissible explanation set and numerical budget; qualify those additions with independent routes and preserved failure records.

3. Study transfer to one measured engineering artifact with calibrated observations and a predeclared claim. Separate model discrepancy and instrument uncertainty from software verification.

10 Reproducibility record

The source snapshot is `da661f54fcd7d6d71aed28cb2b1e4384fdd98aa8`; the retained portfolio pin is `0381c1d2bfe68471494b0710a3fc81e00ae2a98a`. Paths and hashes in the companion manifest pin the exact evidence used. Historical generated tables retain their original statuses; the admission ledger is the authority for current review disposition. This report does not merge those dates into an invented single execution.

Table 7: Reproduction modules and evidence roles, relative to the source repository. Commands describe retained workflows; they were not rerun to create this report.

Python module / artifact	Role
`studies.evidence_sufficiency.run`	Seeded axial artifacts and comparator smoke test
`studies.evidence_sufficiency.claims.run`	Finite claim-specific sufficiency witness
`studies.reference_qualification.run`	Reference routes and public-target comparison
`studies.development_compute.run`	Expanded deterministic sweep and detectability
`studies.cylinder_convergence.run`	Prospectively selected cylinder resolution
`studies.kirsch_reference.run`	Exact-restriction route and boundary checks
`reports/reference-admission/README.md`	Current scoped admission ledger
`reports/fea-interpretation.json`	Retained FE completion/reference/stability fields

References

[1] M. Ulziibayar, *Engineering evidence sufficiency*, repository and retained artifacts, commit da661f54fcd7, 2026. `https://github.com/500ft/evidence-sufficiency-benchmark`.

[2] D. Golovin, A. Krause and D. Ray, "Near-optimal Bayesian active learning with noisy observations," *Advances in Neural Information Processing Systems 23*, 2010. `https://proceedings.neurips.cc/paper_files/paper/2010/hash/1e6e0a04d20f50967c64dac2d639a577-Abstract.html`.

[3] K. Salari and P. Knupp, *Code Verification by the Method of Manufactured Solutions*, Sandia report SAND2000-1444, 2000. DOI: 10.2172/759450. Primary report archive: `https://digital.library.unt.edu/ark:/67531/metadc702130/`.

[4] Ansys, *Mechanical APDL Verification Manual, VM25: Stresses in a Long Cylinder*, Release 2025 R2. `https://ansyshelp.ansys.com/public/Views/Secured/corp/v252/en/ans_vm/Hlp_V_VM25.html`.

[5] Ansys, *Mechanical APDL Verification Manual, VM92: Insulated Wall Temperature*, Release 2025 R2. `https://ansyshelp.ansys.com/public/Views/Secured/corp/v252/en/ans_vm/Hlp_V_VM92.html`.

[6] OpenGeoSys, *Solving Kirsch's problem using the release nodal force approach*, v6.5.7 benchmark. `https://www.opengeosys.org/6.5.7/docs/benchmarks/small-deformations/kirsch/`.